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CS-499
Enhancement One Narrative

The artifact used for this enhancement is a mobile application that was created in a course I previously took at SNHU, CS-360 Mobile Architecture and Programming. It is an inventory management application developed in Java that supports persistent data for users and items using the Room database. The app allows users to create an account, and then log in. They will then navigate to the inventory screen, where they can view existing inventory items, as well as add, edit, and delete items.

I selected this item for the artifact because the application was very basic in its original form, providing a lot of opportunity to improve it. The item also represents a complete application in the sense that there are multiple connected parts that provide value that could be used in real-life, rather than just a coding exercise. Converting the project to Kotlin provided a way to modernize the application, as Kotlin is the standard programming language for Android applications. This artifact showcased my skills in a couple different aspects. It allowed me to demonstrate skills in Android architecture, including using activities for screen behavior, Room entities for database tables, DAO interfaces for database queries, repositories for data access, and an adapter for displaying inventory items in a RecyclerView. I was also able to demonstrate my ability to take an existing application and identify areas for improvement after analyzing the existing Java code. The artifact was improved by converting the Java classes to Kotlin classes, which reduced boilerplate and improved readability. Kotlin features such as null safety, data classes, lambda expressions were used to make the code less prone to errors. The input validation was also improved. I added input validation to prevent duplicate and blank usernames, blank item names, and negative item quantities. I also added stronger password requirements to improve the baseline security for accounts.

I feel I did meet the course outcomes I had planned to meet for this enhancement. The enhancement strongly aligns with course outcome four because it utilizes modern Android tools and computing practices that will directly provide real-life value in inventory tracking. The enhancement improved the prospect of collaboration because it led to code that is more readable and easier to maintain. If other developers were hypothetically working on the application these improvements would improve the collaborative team environment.

This enhancement provided a great opportunity to learn about the Kotlin programming language and patterns. I gained a better understanding of how Kotlin null safety works. I had to think more carefully about which values could potentially be null values and where it is appropriate to use the “?” operator to allow null values. For example, the Room query that searched for a user needed to potentially return a null UserEntity object if no user was found. Before the enhancement, the code used Java Consumer patterns to return data. Changing those to Kotlin function types really cleaned up the code and made it easier to read. One challenge I had was not breaking the app when changing the code over to Kotlin. Several times I had made changes that would make the app crash in certain spots.